

In the microcanonical ensemble we considered a system that is thermally isolated from the rest of the world. Most thermodynamic systems that are of interest to us are not so well isolated – they are in contact with an environment (which we refer to as a heat bath or reservoir) that is much larger than the system of interest and can exchange energy with it (Fig. 4.1). We will not concern ourselves with the microscopic details of the bath, but assume that there are sufficiently many degrees of freedom in the bath that any amount of energy shifted to the system is negligible compared to the total energy in the bath and that there are sufficiently weak interactions between the bath and the system that they can be treated as distinct. We take the system of interest to be in thermodynamic equilibrium with the bath, which means that the temperature T must be fixed, but the energy of the system, E , can fluctuate about its mean value $U = \langle E \rangle$. As we have seen earlier, the probability distribution for U is such that the most probable values of E are very close to U . We can treat the combination “bath + system of interest” as closed, with total energy U_{Total} , and assume that the mean energy in the system of interest $U \ll U_{\text{Total}}$.

4.1 Partition Function

To connect the statistical properties of the canonical ensemble to thermodynamics, we use a similar approach to the one that we used for the microcanonical ensemble. In the microcanonical ensemble we used the principle of maximum entropy and maximized the Gibbs entropy subject to the constraint on the sum of the probabilities $\{p_j\}$ of the microstates. We found that all microstates were equally probable, but did not assume this in our argument for the form of the Gibbs entropy. In the canonical ensemble there is an additional constraint that we need to consider, that the mean energy of the system has a well-defined value U (we showed in Chapter 2 that the equilibrium energy U is centred around a highly probable value). To take this into account in the same framework as before, we need to introduce a Lagrange multiplier β in addition to λ (introduced in Section 2.2) to capture that the average energy is a conserved quantity. We then maximize the Gibbs entropy (up to a factor of k_B , which does not affect our results for the p_j)

$$S = - \sum_j p_j \ln p_j - \lambda \left(\sum_j p_j - 1 \right) - \beta \left(\sum_j \epsilon_j p_j - U \right), \quad (4.1)$$

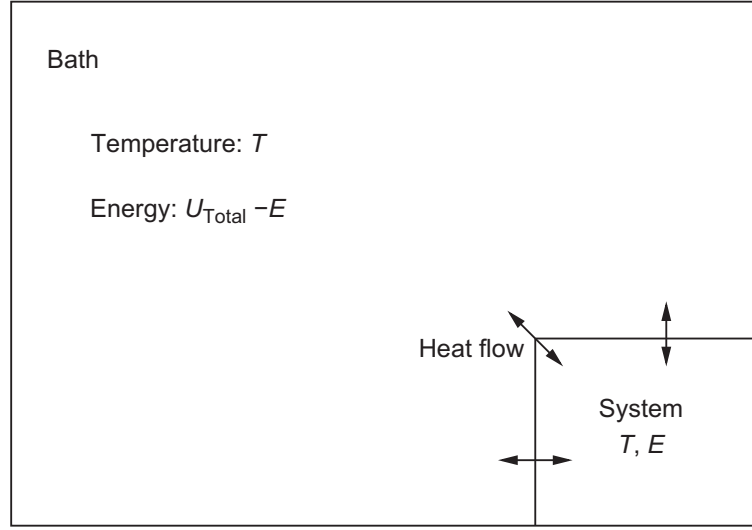


Fig. 4.1 Schematic of the canonical ensemble: system plus heat bath.

where ϵ_j is the energy of microstate j . Similarly to the derivation of the bridge equation for the microcanonical ensemble, we require that the $\{p_j\}$ are chosen to maximize $\mathcal{S}$ which leads to

$$-(\ln p_j + 1) - \lambda - \beta \epsilon_j = 0, \quad (4.2)$$

which we can rearrange to give

$$p_j = e^{-(1+\lambda) - \beta \epsilon_j}. \quad (4.3)$$

Summing over all states on both sides of the equation and using the constraint $\sum_j p_j = 1$ gives

$$1 = e^{-(1+\lambda)} \sum_j e^{-\beta \epsilon_j}, \quad (4.4)$$

and hence we define

$$e^{(1+\lambda)} = \sum_j e^{-\beta \epsilon_j} = Z, \quad (4.5)$$

which fixes λ and introduces the **partition function** as

$$Z = \sum_j e^{-\beta \epsilon_j}. \quad (4.6)$$

Equations (4.3) and (4.5), taken together, imply that

$$p_j = \frac{e^{-\beta \epsilon_j}}{Z}, \quad (4.7)$$

and so we can see that the role of the partition function is to ensure that $\sum_j p_j = 1$. We note that Eq. (4.7) reduces to the result we found in the microcanonical ensemble if all microstates have the same energy ϵ , since in that case

$$Z = \Omega e^{-\beta\epsilon}, \quad (4.8)$$

and then $p_j = 1/\Omega$. At this stage β is a Lagrange multiplier, and we will comment on its physical significance shortly.

We notice that in the canonical ensemble, the probability of a microstate depends on its energy. The partition function ensures that these probabilities are properly normalized. We will see below that its usefulness comes from its relation to thermodynamic properties of the system.

4.2 Bridge Equation in the Canonical Ensemble

In the microcanonical ensemble we found a bridge equation connecting a thermodynamic variable, entropy, with a statistical concept, the multiplicity of states. We can find a bridge equation in the canonical ensemble by starting with the entropy and substituting the expression for the probability of microstate j , Eq. (4.7):

$$\begin{aligned} S &= -k_B \sum_j p_j \ln p_j \\ &= -k_B \sum_j \frac{e^{-\beta\epsilon_j}}{Z} \ln \left(\frac{e^{-\beta\epsilon_j}}{Z} \right) \\ &= -k_B \sum_j \frac{e^{-\beta\epsilon_j}}{Z} (-\beta\epsilon_j) + k_B \sum_j \frac{e^{-\beta\epsilon_j}}{Z} \ln Z \\ &= k_B \beta \sum_j \frac{\epsilon_j e^{-\beta\epsilon_j}}{Z} + k_B \ln Z \sum_j \frac{e^{-\beta\epsilon_j}}{Z} \\ &= k_B \beta U + k_B \ln Z, \end{aligned} \quad (4.9)$$

where we note

$$U = \sum_j \epsilon_j p_j = \frac{\sum_j \epsilon_j e^{-\beta\epsilon_j}}{Z}, \quad (4.10)$$

which is a weighted average of the energies. We still have not determined β . We know that it can be varied so that we get the correct value of the mean energy for our system, but we have not given it a physical meaning.

To determine β , suppose there is a small reversible change in the average energy of the system δU , due to heat flowing into the system from the bath. This implies that there will be a change δp_j in the probability p_j of microstate j , so that we get a new set of probabilities $\{\tilde{p}_j\}$:

$$p_j \rightarrow \tilde{p}_j = p_j + \delta p_j,$$

and

$$\sum_j \tilde{p}_j = 1 \implies \sum_j (p_j + \delta p_j) = 1,$$

which, when combined with the condition $\sum_j p_j = 1$, implies that

$$\sum_j \delta p_j = 0, \quad (4.11)$$

which we will use several times below. Now, the change in energy leads to a change in the entropy of the system, δS , which we can calculate as follows:

$$\begin{aligned} \delta S &= S(\{\tilde{p}_j\}) - S(\{p_j\}) \\ &= -k_B \sum_j (p_j + \delta p_j) \ln(p_j + \delta p_j) + k_B \sum_j p_j \ln p_j \\ &= -k_B \sum_j p_j \ln(p_j + \delta p_j) - k_B \sum_j \delta p_j \ln(p_j + \delta p_j) + k_B \sum_j p_j \ln p_j, \end{aligned} \quad (4.12)$$

and

$$\ln(p_j + \delta p_j) = \ln p_j + \ln\left(1 + \frac{\delta p_j}{p_j}\right), \quad (4.13)$$

hence the second term can be expanded to lowest order in δp_j by using

$$\ln(1 + x) = x + \dots, \quad (4.14)$$

to give

$$\begin{aligned} \delta S &= -k_B \sum_j \delta p_j \ln p_j - k_B \sum_j \delta p_j \\ &= k_B \sum_j \delta p_j (\beta \epsilon_j + \ln Z) \\ &= k_B \beta \sum_j \epsilon_j \delta p_j + k_B \ln Z \sum_j \delta p_j \\ &= k_B \beta \delta U. \end{aligned} \quad (4.15)$$

Hence in the limit of infinitesimal δU we obtain

$$\frac{\delta S}{\delta U} \rightarrow \left. \frac{\partial S}{\partial U} \right|_{N,V} = k_B \beta = \frac{1}{T}, \quad (4.16)$$

where we recalled Eq. (2.11), which allows us to identify

$$\beta = \frac{1}{k_B T}. \quad (4.17)$$

We can now rewrite our equation involving S , U and Z as

$$TS = U + k_B T \ln Z,$$

or

$$U - TS = -k_B T \ln Z, \quad (4.18)$$

and $U - TS = F$, the Helmholtz free energy, so our bridge equation is

$$F = -k_B T \ln Z. \quad (4.19)$$

The equilibrium state may be found by minimizing the Helmholtz free energy (equivalent to maximizing the entropy with constraints) as we are used to in thermodynamics.

In our arguments above, we wrote expressions in terms of U . However, in practice, we usually do not know U and measure the temperature T from which the mean energy can be inferred. This is the way we will approach calculations.

4.2.1 Boltzmann Distribution

With the identification that $\beta = 1/k_B T$, an implication for the probabilities p_j that we found above is that if we compare the probabilities of two states 1 and 2 with energies ϵ_1 and ϵ_2 , respectively, then we get

$$\frac{p(\epsilon_1)}{p(\epsilon_2)} = \frac{\exp\left\{-\frac{\epsilon_1}{k_B T}\right\}}{\exp\left\{-\frac{\epsilon_2}{k_B T}\right\}} = \exp\left\{-\frac{(\epsilon_1 - \epsilon_2)}{k_B T}\right\}, \quad (4.20)$$

which is known as the Boltzmann distribution, with the individual exponentials

$$\exp\left\{-\frac{\epsilon}{k_B T}\right\},$$

known as Boltzmann factors. The probabilities determined in Eq. (4.7) are normalized Boltzmann factors, and exploiting the relationship between the partition function Z and the Helmholtz free energy we may write them as

$$p(\epsilon_j) = \exp\left\{\frac{F - \epsilon_j}{k_B T}\right\}. \quad (4.21)$$

4.2.2 Derivatives of the Partition Function

Once we know the Helmholtz free energy, there are many thermodynamic quantities we can determine through use of partial derivatives, making use of our knowledge of thermodynamics. These can also be related directly to derivatives of the partition function Z , as we see below.

The mean energy is

$$U = \langle E \rangle = \sum_j \epsilon_j p_j = \frac{\sum_j \epsilon_j e^{-\beta \epsilon_j}}{Z}, \quad (4.22)$$

which can be written in the form

$$U = -\frac{1}{Z} \frac{\partial Z}{\partial \beta} = -\frac{\partial \ln Z}{\partial \beta} = k_B T^2 \frac{\partial \ln Z}{\partial T} = k_B T^2 \frac{1}{Z} \frac{\partial Z}{\partial T}, \quad (4.23)$$

where we noted

$$\frac{\partial}{\partial \beta} = \frac{\partial T}{\partial \beta} \frac{\partial}{\partial T} = \frac{1}{\frac{\partial \beta}{\partial T}} \frac{\partial}{\partial T} = -k_B T^2 \frac{\partial}{\partial T}. \quad (4.24)$$

We can evaluate the heat capacity in a similar manner:

$$\begin{aligned} C_V &= \left. \frac{\partial U}{\partial T} \right|_V \\ &= \frac{\partial}{\partial T} \left[\frac{1}{Z} \sum_j \epsilon_j e^{-\beta \epsilon_j} \right] \\ &= \frac{1}{Z} \sum_j \frac{\epsilon_j^2}{k_B T^2} e^{-\beta \epsilon_j} - \frac{1}{Z^2} \frac{\partial Z}{\partial T} \sum_j \epsilon_j e^{-\beta \epsilon_j} \\ &= \frac{1}{k_B T^2} \langle E^2 \rangle - \frac{\partial \ln Z}{\partial T} \frac{1}{Z} \sum_j \epsilon_j e^{-\beta \epsilon_j} \\ &= \frac{1}{k_B T^2} (\langle E^2 \rangle - \langle E \rangle^2), \end{aligned} \quad (4.25)$$

where we used Eq. (4.23) to relate the average energy U to a derivative of the partition function.

Equation (4.25) is an example of a relation between a macroscopic susceptibility (C_V , the linear response of the energy to a temperature change) and equilibrium microscopic fluctuations of the energy. This is actually very general behaviour in equilibrium and is a particular example of the *fluctuation–dissipation theorem*. The significance of the result in this particular example is that it means one can determine the heat capacity of a system without ever changing its temperature, simply by observing equilibrium fluctuations in the energy.

Entropy and Pressure

We already have an expression for the entropy in terms of the multiplicity function Ω , but if we want to calculate the entropy in the canonical ensemble, then it is useful to know the following expression:

$$S = - \left. \frac{\partial F}{\partial T} \right|_{N,V}, \quad (4.26)$$

which we can check easily using the bridge equation:

$$\begin{aligned} -\frac{\partial F}{\partial T} &= \frac{\partial}{\partial T} (k_B T \ln Z) \\ &= k_B \ln Z + \frac{k_B T}{Z} \frac{\partial Z}{\partial T} \end{aligned}$$

$$\begin{aligned}
&= -\frac{F}{T} + \frac{U}{T} \\
&= S.
\end{aligned} \tag{4.27}$$

Another quantity of interest that has not arisen in the examples we have considered so far is that of pressure (obviously of interest for discussing gases). This can be obtained from partial derivatives of either the internal energy (see Eq. (B.18)):

$$P = - \left. \frac{\partial U}{\partial V} \right|_{N,S}, \tag{4.28}$$

or the Helmholtz free energy (see Eq. (B.32)):

$$P = - \left. \frac{\partial F}{\partial V} \right|_{N,T}. \tag{4.29}$$

We can substitute Eq. (4.19) into Eq. (4.29) and then

$$\begin{aligned}
P &= \frac{\partial}{\partial V} k_B T \ln Z \\
&= k_B T \frac{1}{Z} \frac{\partial Z}{\partial V} \\
&= k_B T \frac{1}{Z} \sum_n \frac{\partial}{\partial V} e^{-\beta \epsilon_n} \\
&= \frac{1}{Z} \sum_n \left(-\frac{\partial \epsilon_n}{\partial V} \right) e^{-\beta \epsilon_n},
\end{aligned} \tag{4.30}$$

so from a statistical mechanical point of view, pressure reflects the dependence of the allowed energies on system size.

4.2.3 Equivalence of the Canonical and Microcanonical Ensembles

We have introduced the statistical ensembles as a way to relate microscopic degrees of freedom to macroscopic thermodynamic degrees of freedom. As such, we would hope that the relations we get for thermodynamic quantities should be independent of the method we use to obtain them. In contrast, it is not obvious that the microcanonical and canonical ensembles should give the same results, since for the former, the energy of the system is fixed, whereas for the latter, the energy of the system can fluctuate. However, by rearranging Eq. (4.25), we can look at relative fluctuations of the energy about the mean in the canonical ensemble:

$$\frac{\Delta E}{\langle E \rangle} = \frac{\sqrt{\langle E^2 \rangle - \langle E \rangle^2}}{\langle E \rangle} = \frac{\sqrt{k_B T^2 C_V}}{\langle E \rangle} \sim \frac{1}{\sqrt{N}}, \tag{4.31}$$

and we see that since C_V and $\langle E \rangle$ are both extensive quantities and scale with N , the relative size of fluctuations tends to zero in the $N \rightarrow \infty$ limit, so the probability of finding the system with an energy other than $\langle E \rangle$ vanishes as $N \rightarrow \infty$. Hence even though the energy is allowed to fluctuate, in practice, the energy in the canonical ensemble for sufficiently

large systems takes a well-defined value and takes the same value as that obtained in the microcanonical ensemble. Thus, averages of thermodynamic quantities in the canonical ensemble are dominated by the same set of states used for averages in the microcanonical ensemble, and will give the same results in the $N \rightarrow \infty$ limit.

4.3 Connections to Thermodynamics

We have already seen that thermodynamics and statistical mechanics are intimately related, but it is worth exploring the connection between the two in more depth. The key physics that comes from statistical mechanics is that it provides a microscopic basis for thermodynamics – using knowledge of the microscopic degrees of freedom, the microstates, one can calculate statistical properties of the system that can then be related to thermodynamic quantities. We have seen this already in the bridge equations that relate statistical and thermodynamic quantities for the microcanonical and canonical ensembles. Thermodynamics is a theory of macrostates without consideration of microscopic degrees of freedom, and statistical mechanics connects the microstates with macrostates. In particular, thermodynamics is a theory for $N \rightarrow \infty$, a limit often referred to as the *thermodynamic limit*. In statistical mechanics we find that as $N \rightarrow \infty$ relative fluctuations in thermodynamic quantities $\sim 1/\sqrt{N}$, e.g. in Eq. (4.31), so thermodynamic variables have well-defined limits when N becomes large. However, statistical mechanics also allows one to focus on systems with finite N , and its methods are applicable to problems in information theory, economics and complexity theory, for instance, where the emergent behaviour in the limit of large N does not have an obvious connection to thermodynamics.

Whilst statistical mechanics provides a microscopic justification for thermodynamics, thermodynamics is itself a logically self-contained theory, based on the three laws, supplemented by the zeroth law. We briefly review each of the laws in light of what we have learned thus far.

- (0) **Zeroth law:** *If two systems are in equilibrium with a third, they are in equilibrium with each other.*

We saw that when two systems are in thermal contact in equilibrium, requiring the energy to be distributed in the most probable way implies that $\frac{\partial S}{\partial U}$ takes a constant value in both systems, which we identify as $1/T$. Thus, the two systems in equilibrium have the same value of temperature. We can imagine bringing the two systems in equilibrium in contact with a third, and the condition for equilibrium would thus be that all three systems have the same temperature, consistent with the statement of the zeroth law.

- (1) **First law:** *Conservation of energy.*

We assume that conservation of energy holds at the microscopic level, which then implies that it must also hold at the macroscopic level. We can argue for this by contradiction – if it didn't hold at the microscopic level, we would have no reason to believe it should hold at the macroscopic level.

- (2) **Second law:** *Entropy always increases.*

When we investigated two systems in thermal contact, we saw that after we removed an internal constraint, the entropy increased because $\Omega_f > \Omega_i$.

- (3) **Third law:** *The entropy per particle goes to zero at absolute zero.*

This follows from the statistical definition of entropy

$$S(T = 0) = k_B \ln \Omega_0, \quad (4.32)$$

where Ω_0 is the ground-state degeneracy. In most quantum systems, the ground-state degeneracy is finite, in which case the entropy per particle S/N goes to zero, affirming the third law. There are systems with macroscopically many low-energy states, such as glasses (although glasses are not in equilibrium) and certain spin systems on special lattices (e.g. an antiferromagnet¹ on a kagome lattice), which have macroscopically many classical ground states (however, such degeneracy is very susceptible to being broken by small perturbations).

4.4 Examples

Having introduced the formalism of the canonical ensemble, we now demonstrate its application in a variety of examples. We will consider one-particle examples for which we will obtain the one-particle partition function Z_1 . Given our previous discussion that the thermodynamic limit applies as $N \rightarrow \infty$, it might seem odd to consider $N = 1$. However, we will see in [Section 4.5.2](#) how to relate these one-particle results to partition functions for systems with $N \gg 1$ and how to obtain results that apply in the thermodynamic limit. The examples we consider, a two-level system, the simple harmonic oscillator, rigid rotors and a particle in a box, will turn out to allow us to give a statistical mechanical description of a gas of polyatomic molecules.

4.4.1 Two-Level System

Consider a particle with two energy levels at 0 and ϵ , as illustrated in [Fig. 4.2](#). This is about as simple a statistical mechanical system as one can have, and is a simple model for defects in solids. The partition function is straightforward to calculate:

$$Z = \sum_{n=0}^1 e^{-\beta \epsilon_n} = 1 + e^{-\beta \epsilon}. \quad (4.33)$$

It is also straightforward to calculate the average energy

$$U = \frac{0 + \epsilon e^{-\beta \epsilon}}{Z} = \frac{\epsilon e^{-\beta \epsilon}}{1 + e^{-\beta \epsilon}}. \quad (4.34)$$

¹ We will discuss antiferromagnets in [Chapter 10](#).

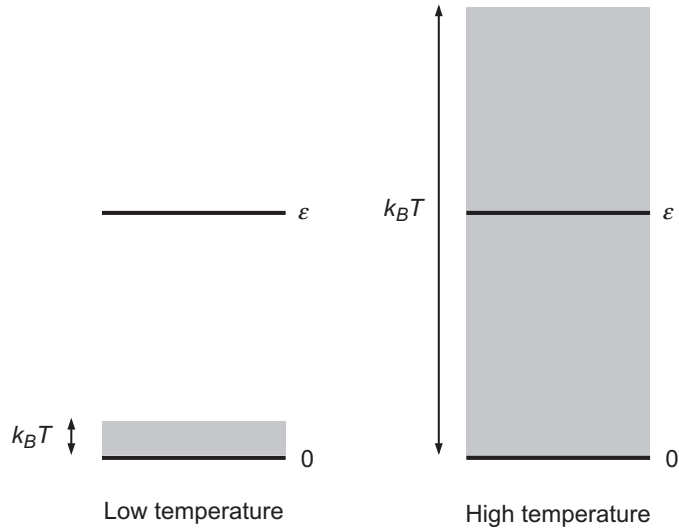


Fig. 4.2 Energy-level diagram for a two-level system. The shaded areas for low and high temperatures indicate energies accessible via thermal fluctuations of order $k_B T$.

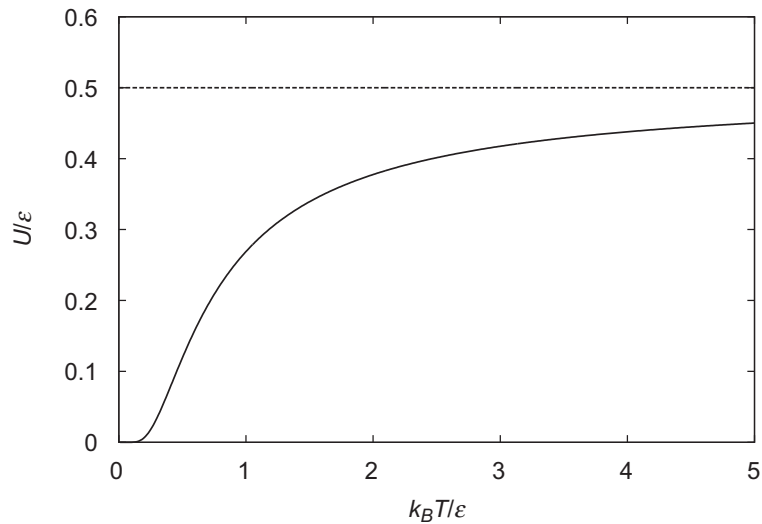


Fig. 4.3 Dimensionless energy as a function of dimensionless temperature for a two-level system.

We plot Eq. (4.34) in Fig. 4.3. We can see that at high temperatures ($\beta \rightarrow 0$), the energy saturates to $U = \epsilon/2$. We can understand this easily by reference to Fig. 4.2. For high temperatures, the thermal energy available is $k_B T \gg \epsilon$, so the difference in the energy of the levels is not an impediment to their occupation. In this limit we expect the probability of the system being in either level to be $\sim 1/2$, and hence the average energy will be $\epsilon/2$. At low temperatures, the probability of excitation to the higher energy level will be

exponentially suppressed, and so the energy will grow very slowly with temperature until $k_B T$ is an appreciable fraction of ϵ and the upper level has non-negligible probability of being occupied.

We can also calculate the heat capacity at constant volume

$$C_V = \left. \frac{\partial U}{\partial T} \right|_V, \quad (4.35)$$

which is

$$\begin{aligned} C_V &= \frac{\partial}{\partial T} \left[\frac{\epsilon e^{-\beta\epsilon}}{1 + e^{-\beta\epsilon}} \right] \\ &= \epsilon \frac{\partial}{\partial T} \left[\frac{1}{1 + e^{\beta\epsilon}} \right] \\ &= \epsilon \left(\frac{\epsilon}{k_B T^2} \right) \frac{e^{\beta\epsilon}}{(1 + e^{\beta\epsilon})^2} \\ &= k_B \left(\frac{\epsilon}{k_B T} \right)^2 \frac{e^{-\beta\epsilon}}{(1 + e^{-\beta\epsilon})^2}, \end{aligned} \quad (4.36)$$

which we can write in terms of a function of a single dimensionless parameter:

$$\frac{C_V}{k_B} = f\left(\frac{\epsilon}{k_B T}\right), \quad (4.37)$$

where

$$f(x) = \frac{x^2 e^{-x}}{(1 + e^{-x})^2}. \quad (4.38)$$

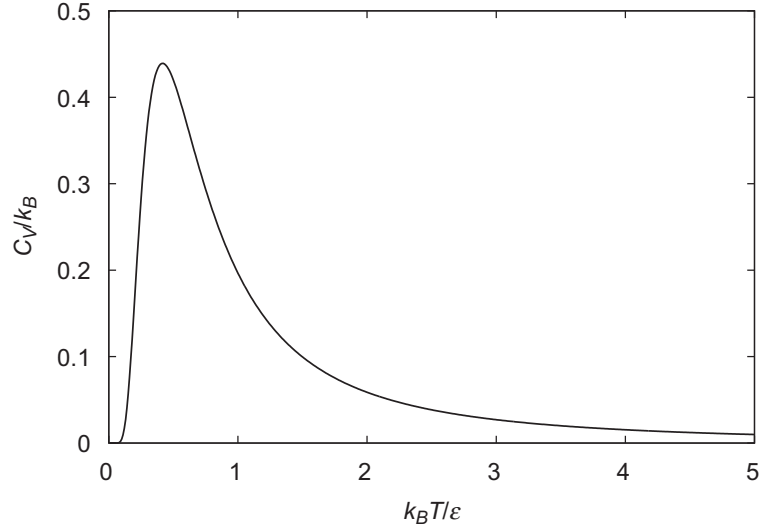
From Eqs (4.37) and (4.38) we can determine several asymptotic limits. At low temperatures, $k_B T \ll \epsilon$, the heat capacity is exponentially small:

$$C_V \sim k_B \left(\frac{\epsilon}{k_B T} \right)^2 e^{-\beta\epsilon}, \quad (4.39)$$

because the only way to change the energy of the system is to increase the occupation probability of the higher level, which is exponentially small at low temperatures due to the presence of an energy gap to the higher level. At high temperatures, $k_B T \gg \epsilon$,

$$C_V \sim k_B \left(\frac{\epsilon}{2k_B T} \right)^2, \quad (4.40)$$

which becomes very small since in the high-temperature limit both states become essentially equally probable, so U changes very slowly with T . The occupation probability of the higher-energy state must always be less than that of the lower-energy state, hence C_V must be very small. We plot the full expression for the heat capacity in Fig. 4.4. In the region between the low- and high-temperature limits, there is a peak which is an example of a Schottky anomaly – a peak in the heat capacity at low temperatures – which occurs in the temperature range between 0 and ϵ , when the probability for excitation to the upper level grows rapidly with temperature.

**Fig. 4.4**

Heat capacity as a function of dimensionless temperature for a two-level system. The peak for $k_B T/\epsilon \sim 0.4$ is an example of a Schottky anomaly.

4.4.2 Quantum Simple Harmonic Oscillator

The energy levels for the quantum simple harmonic oscillator (SHO) are given by the eigenvalues of the simple harmonic oscillator Hamiltonian:

$$H = \frac{p^2}{2m} + \frac{1}{2}kx^2, \quad (4.41)$$

and take the values

$$\epsilon_j = \hbar\omega \left(j + \frac{1}{2} \right) \quad (4.42)$$

for integers $j \geq 0$ with $\omega = \sqrt{k/m}$. Hence the corresponding probability for the state labelled j is

$$p_j = \frac{e^{-\beta\hbar\omega(j+\frac{1}{2})}}{Z}, \quad (4.43)$$

and the partition function can be calculated by noting that it has the form of an infinite geometric series:

$$\begin{aligned} Z &= \sum_{j=0}^{\infty} e^{-\beta\epsilon_j} \\ &= \sum_{j=0}^{\infty} e^{-\beta\hbar\omega(j+\frac{1}{2})} \\ &= e^{-\frac{\beta\hbar\omega}{2}} \sum_{j=0}^{\infty} e^{-j\beta\hbar\omega} \end{aligned}$$

$$\begin{aligned}
&= \frac{e^{-\frac{\beta\hbar\omega}{2}}}{1 - e^{-\beta\hbar\omega}} \\
&= \frac{1}{e^{\frac{\beta\hbar\omega}{2}} - e^{-\frac{\beta\hbar\omega}{2}}} \\
&= \frac{1}{2 \sinh\left(\frac{\hbar\omega}{2k_B T}\right)}, \tag{4.44}
\end{aligned}$$

where we recalled that

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}, \tag{4.45}$$

for $|x| < 1$. Once we have the partition function we can calculate all other thermodynamic quantities of interest. If we are interested in the average energy U then we can obtain it by taking derivatives of the partition function, as we showed in Eq. (4.23), or by direct calculation. We will demonstrate a direct calculation here, but note that it is simpler to use $U = -\frac{1}{Z} \frac{\partial Z}{\partial \beta}$. We start from the definition of U and then we can calculate that

$$\begin{aligned}
U &= \sum_{j=0}^{\infty} \epsilon_j p_j = \hbar\omega \sum_{j=0}^{\infty} \frac{\left(j + \frac{1}{2}\right) e^{-\beta\hbar\omega(j+\frac{1}{2})}}{Z} \\
&= \hbar\omega \left[\frac{e^{-\frac{\beta\hbar\omega}{2}}}{Z} \sum_{j=0}^{\infty} j e^{-j\beta\hbar\omega} + \frac{1}{2} \right], \tag{4.46}
\end{aligned}$$

and we can use the trick that

$$\begin{aligned}
\sum_{j=0}^{\infty} j e^{-j\beta\hbar\omega} &= - \left[\frac{\partial}{\partial x} \sum_{j=0}^{\infty} e^{-jx} \right]_{x=\beta\hbar\omega} \\
&= - \frac{\partial}{\partial x} \left[\frac{1}{1 - e^{-x}} \right]_{x=\beta\hbar\omega} \\
&= \left[\frac{e^{-x}}{(1 - e^{-x})^2} \right]_{x=\beta\hbar\omega}. \tag{4.47}
\end{aligned}$$

Hence, making use of the expression we found for Z in deriving Eq. (4.44), we obtain

$$\begin{aligned}
U &= \frac{\hbar\omega}{2} + \frac{\hbar\omega e^{-\beta\hbar\omega}}{1 - e^{-\beta\hbar\omega}} \\
&= \frac{\hbar\omega}{2} + \frac{\hbar\omega}{e^{\beta\hbar\omega} - 1} \\
&= \frac{\hbar\omega}{2} \left[\frac{e^{\beta\hbar\omega} + 1}{e^{\beta\hbar\omega} - 1} \right] \\
&= \frac{\hbar\omega}{2} \coth\left(\frac{\hbar\omega}{2k_B T}\right). \tag{4.48}
\end{aligned}$$

If we view the energy of the quantum SHO as determined by exciting quanta of an oscillator, so that an energy ϵ_n corresponds to exciting n quanta with energy $\hbar\omega$, then the mean number of excitations in the mode will thus be given by

$$\langle n_{\hbar\omega} \rangle = \frac{U - \frac{1}{2}\hbar\omega}{\hbar\omega} = \frac{1}{e^{\beta\hbar\omega} - 1}. \quad (4.49)$$

As we shall see in [Chapter 9](#), this formula also arises in the context of calculating Planck's formula for the energy density of blackbody radiation, and can be related to the mean occupation of photon modes in a cavity.

4.4.3 Classical Partition Function and Classical Harmonic Oscillator

The SHO gives an example where we can compare the result obtained from the discrete spectrum of a quantum problem to that obtained from the continuous spectrum of the equivalent classical problem. In writing down the partition function for the classical harmonic oscillator, we start from the energies

$$E(p, x) = \frac{p^2}{2m} + \frac{1}{2}kx^2. \quad (4.50)$$

The microstates of the one-dimensional SHO are given by points in two-dimensional classical phase space, and so we replace a discrete sum by an integration over phase space and define the classical partition function by an integral over phase space:

$$Z_{\text{classical}} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{dp dx}{2\pi\hbar} e^{-\beta E(p, x)}. \quad (4.51)$$

The integration over phase space is the classical equivalent of a sum over discrete energy levels in a quantum system. It might seem strange that we have introduced the factor $2\pi\hbar = h$ in the integral.² There are, however, two important reasons for this: (i) it makes the integral dimensionless and (ii) for consistency with the quantum problem (we will show this below). The factor $1/h = 1/(2\pi\hbar)$ has no consequence for calculating averages of physical quantities, since when we calculate averages there will be a factor in the numerator as well as the denominator (Z) and the factors of h will cancel.

When we insert the expression for the energy of the SHO, [Eq. \(4.50\)](#), into the classical partition function [Eq. \(4.51\)](#), we get

$$Z_{\text{classical}} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{dp dx}{2\pi\hbar} e^{-\frac{\beta p^2}{2m}} e^{-\frac{\beta k x^2}{2}}. \quad (4.52)$$

We can perform the Gaussian integrals immediately using [Eq. \(A.2\)](#) to get

$$Z_{\text{classical}} = \frac{1}{2\pi\hbar} \sqrt{2\pi k_B T m} \sqrt{\frac{2\pi k_B T}{k}} = \frac{k_B T}{\hbar\omega}. \quad (4.53)$$

To compare with the quantum problem, we take the classical limit of Z_{quantum} , which corresponds to the case where the available thermal energy is much larger than the separation between adjacent oscillator levels, $k_B T \gg \hbar\omega$, i.e. $\beta\hbar\omega \ll 1$, in which case

² We include a factor of $2\pi\hbar$ for each pair $dp dx$, so for example in three dimensions where we integrate over $d^3\mathbf{p} d^3\mathbf{x}$, there is a factor of $(2\pi\hbar)^3$.

$$\begin{aligned}
Z_{\text{quantum}} &= \frac{e^{-\frac{\beta\hbar\omega}{2}}}{1 - e^{-\beta\hbar\omega}} \\
&\rightarrow \frac{1}{1 - (1 - \beta\hbar\omega + \dots)} \\
&= \frac{k_B T}{\hbar\omega}.
\end{aligned} \tag{4.54}$$

We see that the classical limit of Z_{quantum} is indeed $Z_{\text{classical}}$, validating the normalization we chose for the classical partition function.

4.4.4 Rigid Rotor

Molecules can have rotational degrees of freedom, and in general the energy eigenvalues associated with rotations depend on the distribution of mass in the molecule. The Hamiltonian for a rigid rotor is

$$H_{\text{rigid rotor}} = \frac{J_a^2}{2I_a} + \frac{J_b^2}{2I_b} + \frac{J_c^2}{2I_c}, \tag{4.55}$$

which depends on the angular momentum components $J_a = \mathbf{J} \cdot \hat{\mathbf{a}}$, $J_b = \mathbf{J} \cdot \hat{\mathbf{b}}$ and $J_c = \mathbf{J} \cdot \hat{\mathbf{c}}$, where $\hat{\mathbf{a}}$, $\hat{\mathbf{b}}$ and $\hat{\mathbf{c}}$ are the principal axes of the molecule, and the principal moments of inertia are I_a , I_b and I_c . In general, when $I_a \neq I_b \neq I_c$, this problem does not have an analytic solution, but analytic expressions for the energies can be found for high-symmetry situations. Three of these higher-symmetry situations are the spherical rotor ($I_a = I_b = I_c$), the symmetric rotor ($I_a = I_b \neq I_c$) and the linear rotor ($I_a = I_b, I_c \rightarrow 0$). We discuss each of these examples in turn below.

When we calculate the partition function and sum over states, we need to count all of the states that have the same energy. In such a situation we can write the partition function in the form

$$Z = \sum_j g_j e^{-\beta\epsilon_j}, \tag{4.56}$$

where g_j is the degeneracy of states with energy ϵ_j . The degeneracy depends on the symmetry of the rotor. While superficially the form of the partition function as written in Eq. (4.56) appears to be different from the form we introduced earlier in Eq. (4.6), their physical content is the same. In Eq. (4.6) we introduced the partition function as a sum over states. However, we can group states with the same energy together, in which case the partition function may be rewritten as Eq. (4.56). It is important to be aware that both forms of the partition function are widely used – we will try to be clear as to which form we are using at any particular time.

Spherical Rotor

The spherical rotor corresponds to a situation in which all three moments of inertia are equal: $I_a = I_b = I_c = I$. Examples of molecules that can be described as spherical rotors

include methane (CH_4), uranium hexafluoride (UF_6) and buckyballs (C_{60}). The energy eigenvalues for a spherical rotor are

$$\epsilon_j^{\text{spherical rotor}} = \frac{\hbar^2}{2I} j(j+1), \quad (4.57)$$

where $j = 0, 1, 2, \dots$ is a non-negative integer. The degeneracy of these levels is $g_j = (2j+1)^2$.

Symmetric Rotor

For the symmetric rotor, two moments of inertia are the same, and one differs (e.g. $I_a = I_b \neq I_c$). Examples of molecules that can be described as symmetric rotors include ammonia (NH_3) and benzene (C_6H_6). In this case the energy eigenvalues are

$$\epsilon_{j,k}^{\text{symmetric rotor}} = \frac{\hbar^2}{2I_a} j(j+1) + \frac{\hbar^2}{2} \left(\frac{1}{I_c} - \frac{1}{I_a} \right) k^2, \quad (4.58)$$

where the solution is now labelled by two integers, j and k . The allowed values of j run over non-negative integers as for the spherical rotor, and in addition, the allowed values of k from $-j, -j+1, \dots, 0, \dots, j-1, j$. Thus for each energy eigenvalue the degeneracy is $g_j = 2(2j+1)$ if $k \neq 0$ and $2j+1$ if $k = 0$.

Linear Rotor

We can view the linear rotor as the $I_c \rightarrow 0$ limit of the symmetric rotor, corresponding e.g. to a diatomic molecule, or a linear molecule such as CO_2 . In this case, the only allowed value of k is $k = 0$, and with $I_a = I_b = I$ the energy eigenvalues are

$$\epsilon_j^{\text{linear rotor}} = \frac{\hbar^2}{2I} j(j+1), \quad (4.59)$$

with a degeneracy of $g_j = 2j+1$. In what follows, we will only consider linear rotors when discussing molecular rotations.

For the linear rotor the partition function is

$$Z = \sum_{j=0}^{\infty} (2j+1) \exp \left\{ -\frac{\hbar^2 j(j+1)}{2Ik_B T} \right\}. \quad (4.60)$$

We can rewrite this as

$$Z = \sum_{j=0}^{\infty} (2j+1) \exp \left\{ -\frac{\theta_r j(j+1)}{T} \right\}, \quad (4.61)$$

where

$$\theta_r = \frac{\hbar^2}{2Ik_B} \quad (4.62)$$

is a characteristic temperature for rotations. We cannot evaluate this partition function precisely, but we can investigate the behaviour in the low- and high-temperature limits.

At low temperatures, $T \ll \theta_r$, only the first few terms in the partition function are non-negligible, so we may approximate

$$Z \simeq 1 + 3 \exp \left\{ -\frac{2\theta_r}{T} \right\}, \quad (4.63)$$

and at high temperatures, $T \gg \theta_r$, the exponential decays very slowly, so the summand varies slowly as a function of j and many terms in the sum contribute; we may approximate the sum by an integral, in which case we obtain

$$\begin{aligned} Z &\simeq \int_0^\infty dj (2j+1) \exp \left\{ -\frac{\theta_r j(j+1)}{T} \right\} \\ &= \left[-\frac{T}{\theta_r} \exp \left\{ -\frac{\theta_r j(j+1)}{T} \right\} \right]_0^\infty \\ &= \frac{T}{\theta_r}. \end{aligned} \quad (4.64)$$

4.4.5 Particle in a Box

The energy levels for a particle in a one-dimensional box of length L are

$$E_n = \frac{\hbar^2 \pi^2}{2m L^2} n^2, \quad (4.65)$$

where n is a positive integer. Similarly, for a three-dimensional cube of side length L , we can solve the Schrödinger equation using separation of variables to find that the allowed energy levels are

$$E_{n_x, n_y, n_z} = \frac{\hbar^2 \pi^2}{2m L^2} (n_x^2 + n_y^2 + n_z^2), \quad (4.66)$$

with n_x, n_y, n_z all positive integers. We can evaluate the partition function as

$$\begin{aligned} Z &= \sum_{n_x, n_y, n_z=1}^\infty \exp \left\{ -\frac{\hbar^2 \pi^2}{2mk_B T L^2} (n_x^2 + n_y^2 + n_z^2) \right\} \\ &= \sum_{n_x, n_y, n_z=1}^\infty \exp \left\{ -\frac{\theta_t}{T} (n_x^2 + n_y^2 + n_z^2) \right\} \\ &= \left(\sum_{n_x=1}^\infty \exp \left\{ -\frac{\theta_t n_x^2}{T} \right\} \right)^3, \end{aligned} \quad (4.67)$$

where we introduced the characteristic temperature for translational motion:

$$\theta_t = \frac{\hbar^2 \pi^2}{2mk_B L^2}, \quad (4.68)$$

and we noted that the sums over n_x, n_y and n_z are identical. In the limit of high temperatures, i.e. $T \gg \theta_t$ (this is actually fairly easily achieved, for example for H_2 gas in a container of size $L = 1 \mu\text{m}$, $\theta_t = 1.2 \mu\text{K}$ and θ_t decreases with increasing box size), we may convert the sums to integrals as we did when considering rotations and hence

$$\begin{aligned}
Z &= \left(\int_0^\infty dn \exp \left\{ -\frac{\theta_t}{T} n^2 \right\} \right)^3 \\
&= \left(\frac{1}{2} \sqrt{\frac{\pi T}{\theta_t}} \right)^3 \\
&= \left(\frac{\pi T}{4\theta_t} \right)^{\frac{3}{2}} \\
&= \left(\frac{mk_B T}{2\pi\hbar^2} \right)^{\frac{3}{2}} V \\
&= n_Q V,
\end{aligned} \tag{4.69}$$

where we wrote $L^3 = V$ and introduced the quantum concentration

$$n_Q = \left(\frac{mk_B T}{2\pi\hbar^2} \right)^{\frac{3}{2}}. \tag{4.70}$$

We can relate the quantum concentration to the particle density corresponding to having a single particle in a cube of side length equal to the thermal de Broglie wavelength

$$\lambda_T \sim \frac{\hbar}{\sqrt{2mk_B T}} \sim \frac{h}{m\tilde{v}}, \tag{4.71}$$

where

$$\tilde{v} = \sqrt{\frac{2\pi k_B T}{m}} \tag{4.72}$$

is a characteristic thermal velocity scale.³ We will see that the quantity n_Q arises often when discussing quantum statistical mechanics and provides a convenient way to define a demarcation between quantum and classical regimes.

4.5 Ideal Gas

In the examples in [Section 4.4](#), we considered single-particle partition functions, but in a gas we need to take into account that there are many particles. Before we can give a statistical mechanical treatment of ideal gases, we need to understand what happens when we have multiple subsystems coupled to a bath.

4.5.1 Uncoupled Subsystems

Consider the situation in which a system is composed of two non-interacting (or more realistically, very weakly interacting) subsystems, A and B , both of which are connected to a heat bath which has temperature T , as illustrated in [Fig. 4.5](#).

³ Another way to see how the thermal de Broglie wavelength arises is to recall that $E = \frac{p^2}{2m}$ and then the corresponding de Broglie wavelength is $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$. If we choose $E \sim k_B T$ then up to factors of 2π , which give a particularly simple form for the partition function, the de Broglie wavelength is λ_T .

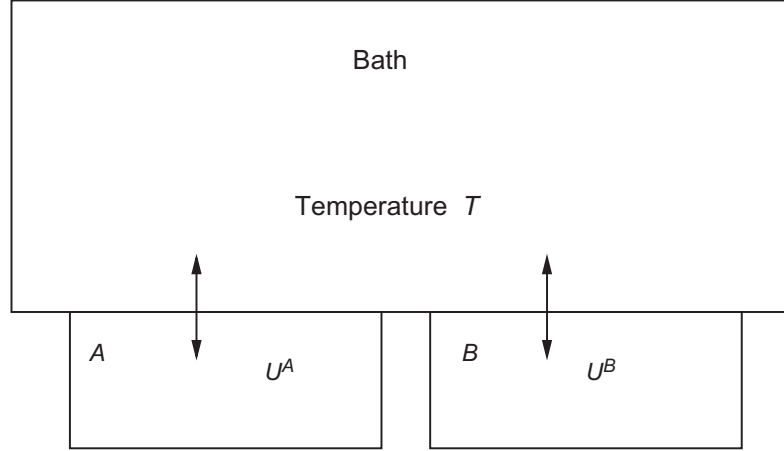


Fig. 4.5

Two non-interacting independent subsystems A and B connected to a bath.

We define the states of subsystem A as s_i^A with energies ϵ_i^A and the states of subsystem B as s_j^B with energies ϵ_j^B , where i and j label the states of A and B , respectively. The states of the system will be pairs (s_i^A, s_j^B) , because the lack of interactions means that the state of A does not affect the state of B and vice versa, and the energy of the state will be $\epsilon_i^A + \epsilon_j^B$,⁴ leading to a partition function for the whole system of

$$\begin{aligned}
 Z &= \sum_{ij} e^{-\beta(\epsilon_i^A + \epsilon_j^B)} \\
 &= \left(\sum_i e^{-\beta\epsilon_i^A} \right) \left(\sum_j e^{-\beta\epsilon_j^B} \right) \\
 &= Z_A Z_B,
 \end{aligned} \tag{4.73}$$

so we see that the partition function factorizes when a system is composed of independent subsystems.

The Helmholtz free energy then takes the form

$$\begin{aligned}
 F &= -k_B T \ln Z \\
 &= -k_B T \ln(Z_A Z_B) \\
 &= -k_B T \ln(Z_A) - k_B T \ln(Z_B) \\
 &= F_A + F_B,
 \end{aligned} \tag{4.74}$$

which is just the sum of the free energies of the independent subsystems. The same is true for other extensive quantities, such as the entropy and average energy.

⁴ In general there may be interactions between two subsystems A and B , in which case the energy of the state would have the form $\epsilon_i^A + \epsilon_j^B + \epsilon_{\text{int};ij}^{AB}$, where $\epsilon_{\text{int};ij}^{AB}$ is an interaction energy.

4.5.2 Distinguishable and Indistinguishable Particles

Whether particles are distinguishable or indistinguishable has consequences for their statistical properties. In order that particles be distinguishable, there needs to be some label that can be used to identify each particle uniquely – for example, particles that sit on fixed sites on a lattice can be distinguished by their position. However, quantum particles such as electrons or protons have no such label and are indistinguishable. We explore the consequences of these differences below.

Distinguishable Particles

From the result for independent subsystems that we established in Eq. (4.73), we can see that if we have N particles that are non-interacting, we can treat each particle as an independent subsystem and the total partition function is a product of the partition functions for the independent particles. Hence if we know the partition function for a single particle Z_1 , then the total partition function is

$$Z = (Z_1)^N, \quad (4.75)$$

if the particles are distinguishable.

We can approach this more formally as follows: if there are n_s particles occupying the energy level ϵ_s , then the total energy for a given set of occupation numbers $\{n_s\}$ is

$$E_{\{n_s\}} = \sum_s n_s \epsilon_s, \quad (4.76)$$

where the n_s have to satisfy the constraint that

$$N = \sum_s n_s. \quad (4.77)$$

The partition function is

$$Z = \sum_{\{n_s\}} g_{\{n_s\}} e^{-\beta E_{\{n_s\}}}, \quad (4.78)$$

where $g_{\{n_s\}}$ is the degeneracy associated with the set of occupation numbers $\{n_s\}$, i.e. the number of different ways we can arrange the distinguishable particles between the energy levels for a given $\{n_s\}$, which is given by

$$g_{\{n_s\}} = \frac{N!}{n_0! n_1! \dots}. \quad (4.79)$$

Then the partition function takes the form

$$\begin{aligned}
 Z &= \sum_{\{n_s\}} \frac{N!}{\prod_s n_s!} e^{-\beta \sum_s n_s \epsilon_s} \\
 &= \sum_{\{n_s\}} \frac{N!}{\prod_s n_s!} \prod_s e^{-\beta n_s \epsilon_s} \\
 &= \left(\sum_s e^{-\beta \epsilon_s} \right)^N \\
 &= (Z_1)^N,
 \end{aligned} \tag{4.80}$$

where we used the multinomial theorem for $k_1 + k_2 + \dots = N$:

$$(x_1 + x_2 + \dots)^N = \sum_{k_1, k_2, \dots} \frac{N!}{k_1! k_2! \dots} x_1^{k_1} x_2^{k_2} \dots, \tag{4.81}$$

in going from the second to the third line to obtain Eq. (4.80).

Indistinguishable Particles

If we consider indistinguishable particles, then for a given set of occupation numbers, there is only one way to arrange the particles. As illustrated in Fig. 4.6, for distinguishable particles we can imagine a label on each of the particles, so the two arrangements of

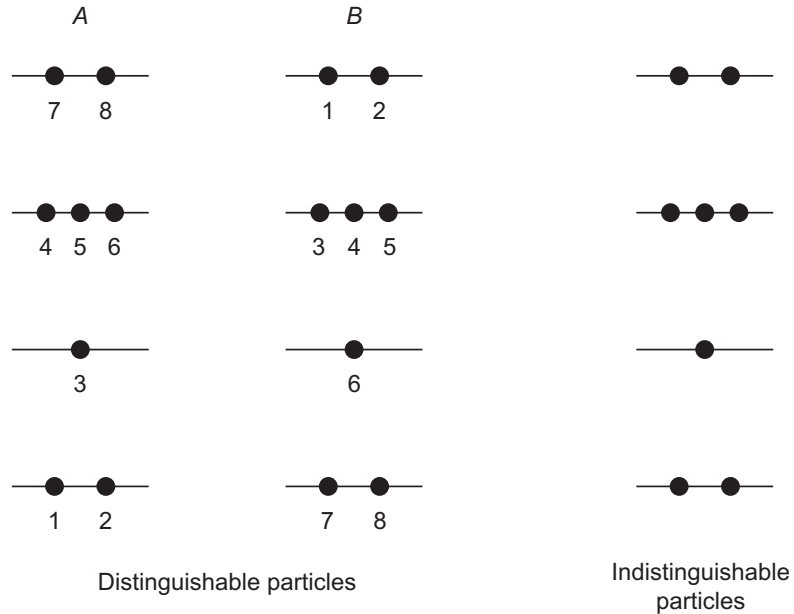


Fig. 4.6

Two arrangements (*A* and *B*) of distinguishable particles (labelled 1 to 8) and one arrangement of indistinguishable particles on a set of energy levels. There are clearly multiple distinct ways to arrange the particles when they are distinguishable, but only one way when they are indistinguishable.

the particles, A and B , are distinct. If the particles are indistinguishable, then A and B correspond to the same arrangement of particles. When we write down the partition function we get

$$Z = \sum_{\{n_s\}} e^{-\beta E_{\{n_s\}}}, \quad (4.82)$$

which cannot be factored in the same way as for distinguishable particles.

The difference between distinguishable and indistinguishable particles becomes very important when we deal with quantum particles (see [Chapter 7](#)), which in addition to being indistinguishable may have further constraints, depending on whether they are bosonic or fermionic. Fermions have half-integer spin, and common examples include electrons, protons, neutrons, neutrinos and ^3He , whereas bosons have integer spin and examples include photons, phonons, magnons and ^4He . An important difference between fermions and bosons is that fermions obey the Pauli exclusion principle, which states that no two fermions can be in the same state.

Example An example which can help to clarify our discussion is when we have two particles that can occupy three available energy levels (1, 2 and 3). If we assume the particles are distinguishable, there are $9 = 3^2$ microstates for the particles, as written out below:

Energy level occupied by each particle

Particle 1:	1	1	1	2	2	2	3	3	3
Particle 2:	1	2	3	1	2	3	1	2	3

For indistinguishable particles, the pairs of occupied levels $\{(1, 2), (2, 1)\}$, $\{(1, 3), (3, 1)\}$ and $\{(2, 3), (3, 2)\}$ are all equivalent, so there are only six possible microstates – this is the number of microstates for indistinguishable bosons. For indistinguishable fermions, the pairs of occupied levels (1, 1), (2, 2) and (3, 3) are forbidden by the Pauli principle, so there are only three possible microstates.

We may rewrite the partition function for indistinguishable particles to reflect the differences between fermions and bosons:

$$Z = \sum_{\{n_s\}}^X e^{-\beta E_{\{n_s\}}}, \quad (4.83)$$

where X is the upper limit of allowed n_s values for a level s and takes the value 1 for fermions and ∞ for bosons. As before, the constraint $\sum_s n_s = N$ holds for the occupation numbers.

This does not seem to make things any easier. However, there is a limit where it is possible to achieve simpler results. If we look at the limit in which there are many more available energy levels than particles, then the probability of double or higher multiple occupancy of a state is miniscule. In this situation, the distinction between fermions and bosons is essentially irrelevant. This is the classical limit and can be realized when the density $n \ll n_Q$, which corresponds to either the low-density or high-temperature limit in

which the interparticle spacing is much larger than the thermal de Broglie wavelength. If there are N particles, then for any microstate, if the particles were distinguishable, there would be $N!$ ways to arrange them, whereas there is only one way to arrange indistinguishable particles, as illustrated in Fig. 4.6. Hence in this limit

$$Z_{\text{indistinguishable}} \simeq \frac{Z_{\text{distinguishable}}}{N!} = \frac{(Z_1)^N}{N!}. \quad (4.84)$$

We will return to examine situations in which the distinctions between fermions and bosons are important later in Chapters 7, 8 and 9.

4.5.3 Ideal Gas

Monatomic Gas

We can apply our discussion of indistinguishable particles to describe a dilute ideal gas. In an ideal gas, the particles are indistinguishable, so that for N atoms, the partition function is given by Eq. (4.84) and we have already calculated Z_1 as the partition function for a particle in a box. If we allow for spin degeneracy (which we didn't in our previous calculation of Z_1), then we should multiply Z_1 by an extra factor g_s and hence we have

$$Z_1 = g_s n_Q V. \quad (4.85)$$

Using Stirling's formula we can rewrite $N! \simeq \sqrt{2\pi N} N^N e^{-N}$, and hence in the large- N limit when we substitute Eq. (4.85) in Eq. (4.84) we get

$$Z = \frac{1}{\sqrt{2\pi N}} \left(\frac{g_s V e n_Q}{N} \right)^N = \frac{1}{\sqrt{2\pi N}} \left(\frac{g_s e n_Q}{n} \right)^N, \quad (4.86)$$

where $n = N/V$ is the number density of atoms. We can use Z to obtain the Helmholtz free energy as

$$F = -k_B T \ln Z = -N k_B T \ln \left(\frac{g_s n_Q}{n} \right) - N k_B T \ln(e) + \frac{k_B T}{2} \ln(2\pi N), \quad (4.87)$$

and we can drop the last term as it is not extensive, so we have

$$F = -N k_B T \ln \left(\frac{g_s n_Q}{n} \right) - N k_B T. \quad (4.88)$$

From here we can calculate all thermodynamic properties of interest, for instance the pressure

$$\begin{aligned} P &= - \left. \frac{\partial F}{\partial V} \right|_T \\ &= \frac{N k_B T}{V}, \end{aligned} \quad (4.89)$$

so we get the ideal gas law

$$PV = N k_B T, \quad (4.90)$$

which can also be written as

$$PV = nRT. \quad (4.91)$$

The form in Eq. (4.91) follows when we recall that Boltzmann's constant is $k_B = 1.3806 \times 10^{-23} \text{ J K}^{-1}$ with N the number of molecules and the ideal gas constant is $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$ with n the number of moles in the gas (1 mole is Avogadro's number $N_A = 6.022 \times 10^{23}$ molecules), so that $Nk_B = nR$. We can also calculate the entropy from

$$\begin{aligned} S &= - \left. \frac{\partial F}{\partial T} \right|_{N,V} \\ &= Nk_B \ln \left(\frac{g_s n_Q}{n} \right) + Nk_B + \frac{3}{2} Nk_B, \end{aligned} \quad (4.92)$$

which gives

$$S = Nk_B \left[\ln \left(\frac{g_s n_Q}{n} \right) + \frac{5}{2} \right], \quad (4.93)$$

which is known as the **Sackur–Tetrode equation** for a monatomic gas, obtained independently by Otto Sackur and Hugo Tetrode in around 1912.

With F and S known, it is straightforward to calculate the energy

$$U = F + TS = \frac{3}{2} Nk_B T, \quad (4.94)$$

and hence the heat capacity at constant volume is

$$C_V = \left. \frac{\partial U}{\partial T} \right|_V = \frac{3}{2} Nk_B. \quad (4.95)$$

We can use Eqs (4.95) and (4.25) to estimate the size of energy fluctuations in an ideal gas and verify our assertion in Section 4.2.3 that in equilibrium, the canonical and microcanonical ensembles are equivalent. From Eqs (4.94) and (4.25) we can estimate the relative size of energy fluctuations:

$$\begin{aligned} \frac{\Delta U}{U} &= \frac{\sqrt{\langle E^2 \rangle - \langle E \rangle^2}}{U} \\ &= \frac{\sqrt{k_B T^2 C_V}}{U} \\ &= \frac{\sqrt{k_B T^2 \frac{3}{2} Nk_B}}{\frac{3}{2} Nk_B T} \\ &= \sqrt{\frac{2}{3N}}. \end{aligned} \quad (4.96)$$

For a mole of ideal gas this gives a fractional fluctuation of $\sim 1.05 \times 10^{-12}$, which is too small to be measured.

Gas of Polyatomic Molecules

For a single polyatomic molecule, there are additional degrees of freedom beyond translations, such as rotations, vibrations and internal electronic transitions. To a good approximation these degrees of freedom can often be treated as independent, so we may write the single-particle partition function as

$$Z_1 = Z_t Z_r Z_v Z_e, \quad (4.97)$$

where Z_t is the single-particle partition function corresponding to translations that we discussed for the monatomic gas. Z_r is the partition function for rotations: for a linear molecule, such as a diatomic molecule, or a triatomic molecule like CO_2 , Z_r would be given by the result we found for a linear rotor in Eq. (4.61), Z_v is the vibrational partition function that might be approximated by a simple harmonic oscillator (at low enough temperatures that anharmonicities can be ignored) and Z_e is the partition function for electronic transitions, which might be modelled by a few-level system for low temperatures (relative to the energy spacing of the transitions). An order of magnitude estimate of the characteristic temperatures for these extra degrees of freedom is $\theta_r \sim 10 - 100$ K, $\theta_v \sim 1000$ K and $\theta_e \sim 10000$ K.

Using the expression for Z_t that we obtained in Eq. (4.85), we can follow the steps to get to Eq. (4.86) and see that

$$Z = \frac{1}{\sqrt{2\pi N}} \left(\frac{g_s n_Q}{n} \right)^N Z_r^N Z_v^N Z_e^N, \quad (4.98)$$

and the Helmholtz free energy is

$$F = -Nk_B T \ln \left(\frac{g_s n_Q}{n} \right) - Nk_B T - Nk_B T \ln(Z_r) - Nk_B T \ln(Z_v) - Nk_B T \ln(Z_e). \quad (4.99)$$

Now, Z_r , Z_v and Z_e are all independent of volume, so we still obtain the ideal gas law

$$PV = Nk_B T \quad (4.100)$$

when we take a derivative of F with respect to V to calculate the pressure. If we are interested in temperatures much less than θ_v and θ_e (which is the case at room temperature, for example) then we can neglect the terms above involving Z_v and Z_e and approximate the entropy as

$$S = Nk_B \left[\ln \left(\frac{g_s n_Q}{n} \right) + \frac{5}{2} \right] + Nk_B \frac{\partial}{\partial T} [T \ln Z_r].$$

At temperatures $T \gg \theta_r$ we know from Eq. (4.64) that $Z_r \simeq \frac{T}{\theta_r}$, in which case we get (neglecting vibrations and electronic excitations)

$$S = Nk_B \left[\ln \left(\frac{g_s n_Q}{n} \frac{T}{\theta_r} \right) + \frac{7}{2} \right], \quad (4.101)$$

and hence the energy is

$$U = \frac{5}{2} Nk_B T, \quad (4.102)$$

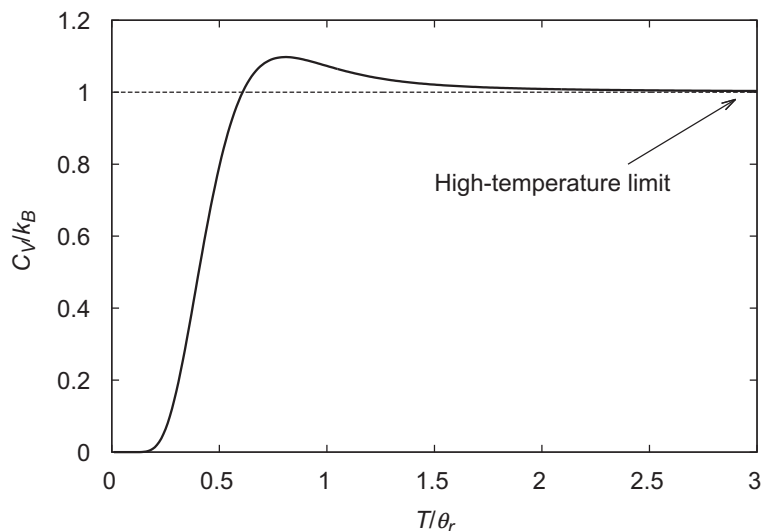


Fig. 4.7

Heat capacity from rotations for a gas of diatomic molecules – note the small peak at a temperature $T \sim 0.8 \theta_r$.

which implies a heat capacity of

$$C_V = \frac{5}{2} N k_B. \quad (4.103)$$

At temperatures much below θ_r , we know that $Z_r \simeq 1 + 3 \exp \{-2\theta_r/T\}$, which will lead to an exponentially small contribution to the heat capacity so that $C_V = \frac{3}{2} N k_B$ as for a monatomic gas. The contribution to the heat capacity from rotations over the full temperature range is plotted in Fig. 4.7.

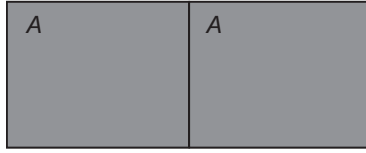
4.5.4 Example: Entropy of Mixing

The factor of $N!$ in Eq. (4.84) has important physical consequences, as first recognized by Gibbs. This is most easily seen in the example of the entropy of mixing. Suppose we have a container with a partition that separates two regions, each with volume V . Consider two scenarios: in scenario 1, both sides are filled with an ideal gas containing N molecules of type A, whereas in scenario 2, one side is filled with an ideal gas containing N molecules of type A and the other contains N molecules of type B, as illustrated in Fig. 4.8. The partition is then removed and then we wait for the gas to equilibrate. We can calculate the change in entropy in each scenario, using the Sackur–Tetrode equation, Eq. (4.93).

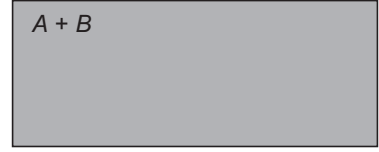
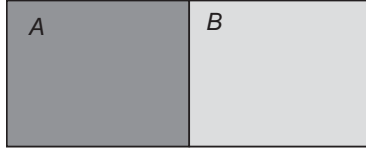
In scenario 1, the initial entropy is given by twice the entropy for N particles in volume V :

$$S_1^{\text{initial}} = 2Nk_B \left[\ln \left(\frac{g_s^A n_Q^A V}{N} \right) + \frac{5}{2} \right], \quad (4.104)$$

Scenario 1



Scenario 2



Initial state

Final state

Fig. 4.8

Two scenarios for entropy of mixing. In scenario 1 there are atoms of type A on both sides of a partition before it is removed and in scenario 2 there are atoms of type A on one side of the partition and atoms of type B on the other side of the partition before it is removed.

where g_s^A is the spin degeneracy and n_Q^A is the quantum concentration for molecules of type A . The final entropy is that of $2N$ particles in a volume $2V$:

$$S_1^{\text{final}} = 2Nk_B \left[\ln \left(\frac{g_s^A n_Q^A 2V}{2N} \right) + \frac{5}{2} \right]. \quad (4.105)$$

Hence, the change in entropy is

$$\Delta S_1 = S_1^{\text{final}} - S_1^{\text{initial}} = 0. \quad (4.106)$$

This is in accord with our expectation that the process of removing the partition is reversible. If we reinsert the partition, then we are back to our original situation in which there are N molecules of type A on each side of the partition – they may not be the same molecules as were there originally, but since they are indistinguishable the situation is equivalent to the original configuration.

In scenario 2, the initial entropy is that of N molecules in volume V for both types of molecules:

$$S_2^{\text{initial}} = Nk_B \left[\ln \left(\frac{g_s^A n_Q^A V}{N} \right) + \frac{5}{2} \right] + Nk_B \left[\ln \left(\frac{g_s^B n_Q^B V}{N} \right) + \frac{5}{2} \right], \quad (4.107)$$

while the final entropy is that of N molecules in volume $2V$ for both types of molecules:

$$S_2^{\text{final}} = Nk_B \left[\ln \left(\frac{g_s^A n_Q^A 2V}{N} \right) + \frac{5}{2} \right] + Nk_B \left[\ln \left(\frac{g_s^B n_Q^B 2V}{N} \right) + \frac{5}{2} \right]. \quad (4.108)$$

Hence the change in entropy is

$$\Delta S_2 = S_2^{\text{final}} - S_2^{\text{initial}} = 2Nk_B \ln 2, \quad (4.109)$$

which is the *entropy of mixing*. The entropy increases, indicating that scenario 2 describes an irreversible process. This is in accord with the observation that if we replace the partition, then the situation is not the same as it was initially – there will be molecules of both type *A* and type *B* on both sides of the partition.

If we review the derivation of the Sackur–Tetrode equation then we see that if the factor of $N!$ had not been included in Eq. (4.84), there would not be a factor of N inside the argument of the logarithm. This would imply that $\Delta S_1 = \Delta S_2$, even though, as we have argued above, one corresponds to a reversible process and the other corresponds to an irreversible process. We will see additional arguments for the presence of $N!$ in the partition function for dilute indistinguishable particles in Chapter 7.

4.6 Non-ideal Gases

We have explored the properties of ideal gases in considerable detail, but ignoring interactions between molecules is an oversimplification. For a more quantitative description of real gases it is necessary to include interactions. We discuss these interactions and show that a simple treatment leads to the van der Waals equation of state.

We consider N classical particles, with the momentum and position of the i^{th} particle labelled $\mathbf{p}_i$ and $\mathbf{r}_i$, respectively. We assume that the particles located at $\mathbf{r}_i$ and $\mathbf{r}_j$ have pairwise interactions via a potential $\phi(|\mathbf{r}_i - \mathbf{r}_j|)$ that depends only on the distance between the particles, $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$. The energy of the system is thus given by

$$E = \sum_i \frac{p_i^2}{2m} + \sum_{j>i} \phi(r_{ij}), \quad (4.110)$$

where the sum is for $j > i$ in the interaction term so that we do not double count any interactions.

A typical interaction potential has the characteristics embodied by the Lennard–Jones potential

$$\phi_{\text{LJ}}(r) = 4\epsilon \left\{ \left(\frac{r_0}{r} \right)^{12} - \left(\frac{r_0}{r} \right)^6 \right\}, \quad (4.111)$$

which is strongly repulsive at short distances ($r < r_0$) and weakly attractive at long distances, as illustrated in Fig. 4.9.⁵

We can write down the N -particle classical partition function

$$Z_N = \frac{g_s^N}{N!} \int \frac{d^3\mathbf{p}_1 \dots d^3\mathbf{p}_N d^3\mathbf{r}_1 \dots d^3\mathbf{r}_N}{(2\pi\hbar)^{3N}} e^{-\beta E}, \quad (4.112)$$

and substituting Eq. (4.110) for the energy E we get

$$Z_N = \frac{g_s^N}{N!} \int \frac{d^3\mathbf{p}_1 \dots d^3\mathbf{p}_N}{(2\pi\hbar)^{3N}} e^{-\beta \sum_i \frac{p_i^2}{2m}} \int d^3\mathbf{r}_1 \dots d^3\mathbf{r}_N e^{-\beta \sum_{j>i} \phi(r_{ij})}. \quad (4.113)$$

⁵ The Lennard–Jones potential is widely used in simulations of fluids as it captures qualitatively important aspects of more realistic interaction potentials.

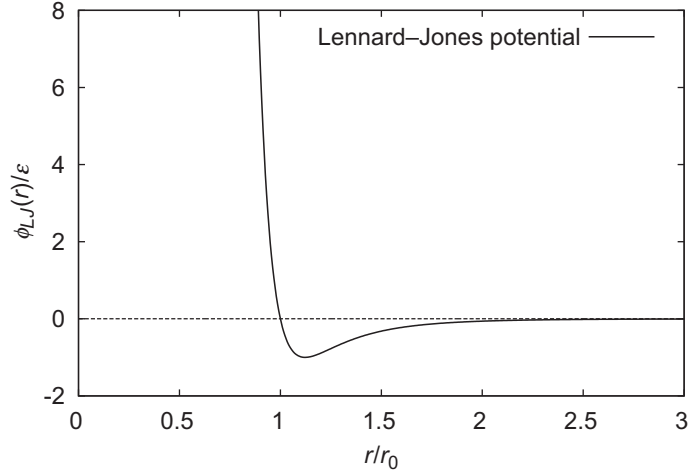


Fig. 4.9

Lennard–Jones potential: note the strongly repulsive short-range piece for $r < r_0$ and the weakly attractive tail for $r > r_0$.

We can perform the momentum integrals to get

$$\left[\int \frac{d^3 \mathbf{p}}{(2\pi\hbar)^3} e^{-\frac{\beta p^2}{2m}} \right]^N = \left(\frac{mk_B T}{2\pi\hbar^2} \right)^{\frac{3N}{2}} = n_Q^N, \quad (4.114)$$

and hence

$$Z_N = \frac{g_s^N n_Q^N}{N!} Z_\phi, \quad (4.115)$$

where

$$Z_\phi = \int d^3 \mathbf{r}_1 \dots d^3 \mathbf{r}_N e^{-\beta \sum_{j>i} \phi(r_{ij})}. \quad (4.116)$$

Now, if $\phi = 0$, we see that $Z_\phi = V^N$ and hence

$$Z_N = \frac{(Z_1)^N}{N!}, \quad \text{with } Z_1 = g_s n_Q V, \quad (4.117)$$

just as we already found for an ideal gas.

In general, Z_ϕ is non-trivial to evaluate. To obtain an approximate form, assume that we can write $\phi(r)$ as the sum of a short-range repulsive piece $\phi_{\text{sr}}(r)$ and a long-range attractive piece $\phi_{\text{lr}}(r)$, similar to the Lennard–Jones potential introduced in Eq. (4.111):

$$\phi(r) = \phi_{\text{sr}}(r) + \phi_{\text{lr}}(r).$$

In this case

$$Z_\phi = \int d^3 \mathbf{r}_1 \dots d^3 \mathbf{r}_N e^{-\beta \sum_{j>i} [\phi_{\text{sr}}(r_{ij}) + \phi_{\text{lr}}(r_{ij})]}, \quad (4.118)$$

and to evaluate this we make the additional approximation for the long-range part of the potential

$$\begin{aligned}\sum_{j>i} \phi_{\text{lr}}(r_{ij}) &= \frac{1}{2} \sum_i \sum_{j \neq i} \phi_{\text{lr}}(r_{ij}) \\ &\simeq \frac{1}{2} \sum_i \frac{(N-1)}{V} \int d^3 \mathbf{r} \phi_{\text{lr}}(r),\end{aligned}\quad (4.119)$$

where we replaced the sum over the potentials of the $N-1$ particles $j \neq i$ by $(N-1)$ times the average value of the potential, which is⁶

$$\frac{1}{V} \int d^3 \mathbf{r} \phi_{\text{lr}}(r).$$

All N contributions to the sum over i will be identical in Eq. (4.119), so in the limit of large N we approximate

$$\sum_{j>i} \phi_{\text{lr}}(r_{ij}) \simeq -\frac{aN^2}{V}, \quad (4.120)$$

where

$$a = -\frac{1}{2} \int d^3 \mathbf{r} \phi_{\text{lr}}(r). \quad (4.121)$$

Thus we can simplify Z_ϕ to

$$Z_\phi \simeq e^{\frac{\beta a N^2}{V}} \int d^3 \mathbf{r}_1 \dots d^3 \mathbf{r}_N e^{-\beta \sum_{j>i} \phi_{\text{sr}}(r_{ij})}. \quad (4.122)$$

Now, for the short-range potential, $\phi_{\text{sr}} \simeq 0$ almost everywhere except when two particles are very close. The repulsive aspect of the short-range potential will have the effect of excluding all of the other $N-1$ particles in some volume around the i^{th} particle. Hence, when $N \gg 1$ we can approximate

$$\int d^3 \mathbf{r}_1 \dots d^3 \mathbf{r}_N e^{-\beta \sum_{j>i} \phi_{\text{sr}}(r_{ij})} \simeq (V - Nb)^N, \quad (4.123)$$

where b is an average effective volume that is excluded by each molecule, leaving the volume that a molecule can actually move around in as effectively $V - Nb$. With these approximations we obtain the partition function

$$Z \simeq \frac{g_s^N}{N!} \left(\frac{mk_B T}{2\pi\hbar^2} \right)^{\frac{3N}{2}} (V - Nb)^N e^{\frac{\beta a N^2}{V}}, \quad (4.124)$$

where the gas is characterized by the parameters a and b . From this approximate partition function it is straightforward (see Problem 4.10) to derive the van der Waals equation of state for a non-ideal gas:

⁶ We saw in Chapter 1 that this approach is valid for random variables due to the central limit theorem, however, here the long-range potential will induce correlations between the particles so that their positions are not truly random variables, and so this is only an approximation.

$$P = \frac{Nk_B T}{V - Nb} - \frac{aN^2}{V^2}. \quad (4.125)$$

The van der Waals equation of state highlights physics left out in the derivation of the ideal gas law – the finite size of molecules and their weak long-range attractions – and while more accurate than the ideal gas law is not generally adequate to give a quantitative description of real gases.

4.7 The Equipartition Theorem

A result that emerges from studying the ideal gas is that the energy is very simple in the classical limit – for the monatomic ideal gas the energy is $\frac{3}{2}Nk_B T$, and in the diatomic gas at temperatures high enough that all rotations are accessible the energy is $\frac{5}{2}Nk_B T$. We notice that in the monatomic gas, the energy is purely kinetic, and there are contributions from motion in the x , y and z directions, which average out to $\frac{1}{2}k_B T$ per particle per degree of freedom. When we allow rotations for a diatomic molecule, we gain two more factors of $\frac{1}{2}k_B T$ per particle in the classical limit. We note that there are only two non-zero moments of inertia for a diatomic molecule, hence there are two degrees of freedom, as we might have guessed from counting multiples of $\frac{1}{2}k_B T$.

The results noted above seem to indicate a pattern, and indeed they can be stated more formally in the **equipartition theorem**:

For a classical system in equilibrium, the average energy of each degree of freedom that the energy is quadratic in is $\frac{1}{2}k_B T$.

To prove this result, suppose that the energy of a system with N quadratic degrees of freedom can be written as

$$E(\{x_i\}) = \sum_{i=1}^N \epsilon_i = \sum_{i=1}^N c_i x_i^2, \quad (4.126)$$

where x_i refers to any co-ordinate of the system (which could be position, momentum, angular momentum, etc.) and the $\{c_i\}$ are a set of constants. Now, the mean energy

$$U = \langle E \rangle = \left\langle \sum_{i=1}^N \epsilon_i \right\rangle = \sum_{i=1}^N \langle \epsilon_i \rangle \quad (4.127)$$

is a sum over the mean energies of each of the co-ordinates, and

$$\langle \epsilon_j \rangle = \int d\Gamma \epsilon_j P(\epsilon_j), \quad (4.128)$$

where $d\Gamma = \prod_{i=1}^N dx_i$. In the canonical ensemble we get

$$\langle \epsilon_j \rangle = \frac{\int d\Gamma \epsilon_j e^{-\beta E(\{x_i\})}}{\int d\Gamma e^{-\beta E(\{x_i\})}}$$

$$\begin{aligned}
&= \frac{\int d\Gamma \epsilon_j e^{-\beta \sum_i \epsilon_i}}{\int d\Gamma e^{-\beta \sum_i \epsilon_i}} \\
&= \frac{\int dx_j \epsilon_j e^{-\beta \epsilon_j} \int d\Gamma' e^{-\beta \sum_{i \neq j} \epsilon_i}}{\int dx_j e^{-\beta \epsilon_j} \int d\Gamma' e^{-\beta \sum_{i \neq j} \epsilon_i}}, \tag{4.129}
\end{aligned}$$

where $d\Gamma' = \prod_{i \neq j}^N dx_i$. Hence, noting that the integrals over $d\Gamma'$ cancel, we find

$$\begin{aligned}
\langle \epsilon_j \rangle &= \frac{\int dx_j \epsilon_j e^{-\beta \epsilon_j}}{\int dx_j e^{-\beta \epsilon_j}} \\
&= \frac{c_j \int_{-\infty}^{\infty} dx_j x_j^2 e^{-\beta c_j x_j^2}}{\int_{-\infty}^{\infty} dx_j e^{-\beta c_j x_j^2}} \\
&= \frac{c_j \frac{\sqrt{\pi}}{2(\beta c_j)^{\frac{3}{2}}}}{\sqrt{\frac{\pi}{\beta c_j}}} \\
&= \frac{1}{2\beta} \\
&= \frac{1}{2} k_B T, \tag{4.130}
\end{aligned}$$

where we used previously established results for Gaussian integrals. The significance of this result is that if we have a system where the energy is quadratic in some variables, then we know their contribution to the energy at high temperatures without having to do any calculation.

4.7.1 Example: The Ideal Gas

For an ideal gas, the energy of a single molecule is

$$E = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m}. \tag{4.131}$$

The energy is quadratic in three variables, p_x , p_y and p_z , so the equipartition theorem tells us that the energy per particle is

$$\frac{E}{N} = \frac{1}{2} k_B T + \frac{1}{2} k_B T + \frac{1}{2} k_B T = \frac{3}{2} k_B T, \tag{4.132}$$

which we have already obtained from a longer calculation.

4.7.2 Dulong and Petit Law

In addition to the ideal gas case considered above, another simple application of the equipartition theorem is to simple harmonic oscillators, where the energy is quadratic in position and momentum. We found that for a single harmonic oscillator

$$U = \frac{\hbar\omega}{2} \coth\left(\frac{\hbar\omega}{2k_B T}\right), \quad (4.133)$$

which in the high-temperature limit, $k_B T \gg \hbar\omega$, becomes

$$U \rightarrow \frac{\hbar\omega}{2} \frac{2k_B T}{\hbar\omega} = k_B T, \quad (4.134)$$

as we would expect from the equipartition theorem for two degrees of freedom.

The Einstein model of a crystal treats the vibrations of a solid (i.e. phonons) as arising from a collection of $3N$ independent harmonic oscillators with the same frequency ω , with the energy given by

$$E = \sum_{i=1}^{3N} \frac{p_i^2}{2m} + \sum_{i=1}^{3N} \frac{1}{2} m \omega^2 x_i^2. \quad (4.135)$$

Hence at high temperatures (room temperature is usually sufficient) the equipartition theorem gives

$$U = \langle E \rangle = 3N \frac{1}{2} k_B T + 3N \frac{1}{2} k_B T = 3N k_B T, \quad (4.136)$$

and hence

$$C_V = \left. \frac{\partial U}{\partial T} \right|_V = 3N k_B. \quad (4.137)$$

For a mole of a material, i.e. $N = N_{\text{Avogadro}} = 6.022 \times 10^{23}$, $C_V = 3R$, where $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ is the universal gas constant.

This provides a microscopic explanation for the Dulong and Petit law that the high-temperature limit of the molar heat capacity of many different crystals is $3R$. The law (not stated in the modern form) was proposed by Pierre Louis Dulong and Alex Thérèse Petit in 1819 based on empirical observations. We will obtain a more complete understanding of the heat capacity of solids when we discuss the Debye model in [Chapter 9](#).

4.8 Summary

In the canonical ensemble, the temperature T , particle number N and volume V are fixed. The equation

$$F = -k_B T \ln Z \quad (4.138)$$

is the bridge equation which relates a thermodynamic concept, the Helmholtz free energy $F = U - TS$, to a statistical mechanical concept, the partition function, which can be written as a sum over states s :

$$Z = \sum_s e^{-\beta \epsilon_s}, \quad (4.139)$$

or a sum over different energy levels ϵ_j , weighted by their degeneracy g_j :

$$Z = \sum_j g_j e^{-\beta \epsilon_j}. \quad (4.140)$$

For classical systems the partition function is expressed as an integral over phase space. For multi-particle systems the form of the partition function depends on whether particles are distinguishable or indistinguishable. For distinguishable particles the N -particle partition function is

$$Z_N = (Z_1)^N, \quad (4.141)$$

where Z_1 is the single-particle partition function and for indistinguishable particles the N -particle partition function is only simple in the dilute limit, in which case it takes the form

$$Z_N \simeq \frac{(Z_1)^N}{N!}. \quad (4.142)$$

Treating an ideal gas as the dilute limit of a gas of indistinguishable particles in a box, for which $Z_1 = g_s n_Q V$, with

$$n_Q = \left(\frac{mk_B T}{2\pi \hbar^2} \right)^{\frac{3}{2}}, \quad (4.143)$$

the quantum concentration, one can derive the ideal gas law

$$PV = Nk_B T, \quad (4.144)$$

and that the energy of an ideal gas is

$$U = \frac{3}{2} Nk_B T. \quad (4.145)$$

The result for the energy can also be obtained from the equipartition theorem, which states that for a classical system in equilibrium, each quadratic degree of freedom contributes $\frac{1}{2} k_B T$ to the energy.

Problems

- 4.1** We can derive the probability that a system is in microstate j in the canonical ensemble by considering the system + reservoir as an isolated system with total energy U , where the reservoir energy is much larger than that of the system. Suppose the system is in a definite microstate j which has energy ϵ_j .

(a) Justify

$$p_j = c \Omega_R(U - \epsilon_j),$$

where c is a constant and Ω_R is the multiplicity function for the reservoir.

(b) Expand $\ln \Omega_R(U - \epsilon_j)$ to show that

$$p_j = \frac{e^{-\beta \epsilon_j}}{Z},$$

where $\beta = 1/k_B T$ and give an expression for Z in terms of c and Ω_R .

- 4.2** We have seen how to calculate C_V for an ideal gas. Calculate C_P for an ideal gas in three dimensions, recalling that it can be written as a derivative of the enthalpy H via

$$C_P = \left. \frac{\partial H}{\partial T} \right|_P = T \left. \frac{\partial S}{\partial T} \right|_P.$$

Relate C_P and C_V for both a monatomic and a diatomic ideal gas.

- 4.3** Estimate θ_r for O_2 and N_2 . You may find it helpful to note that the bond length for oxygen is 1.208 Å and the bond length for nitrogen is 1.0975 Å, and the atomic masses of oxygen and nitrogen are 15.999 amu and 14.0067 amu, respectively.

- 4.4** We saw in Eq. (4.74) that for a system composed of two independent subsystems A and B , the Helmholtz free energy takes the form

$$F = F_A + F_B,$$

which is just the sum of the free energies of the independent subsystems. Show that the same is true for other extensive quantities such as the entropy and average energy.

- 4.5** In Section 4.2.1 we found an expression for the relative probabilities of states with different energies. We can find the same result in a different way:

- (a) The ratio of the probabilities of the system having energies ϵ_1 and ϵ_2 can be related to the number of states that the reservoir has with energies $U - \epsilon_1$ and $U - \epsilon_2$. Use an argument of this form to show that the ratio of probabilities is

$$\frac{P(\epsilon_1)}{P(\epsilon_2)} = e^{\frac{1}{k_B} [S(U - \epsilon_1) - S(U - \epsilon_2)]}.$$

- (b) Taylor expand the right-hand side of the expression in part (a) to show that

$$\frac{P(\epsilon_1)}{P(\epsilon_2)} = \exp \left\{ -\frac{(\epsilon_1 - \epsilon_2)}{k_B T} \right\}.$$

- (c) We showed that the relative fluctuations in the energy vanish in the limit $N \rightarrow \infty$. Show that the probability that the system has an energy $E = \langle E \rangle + \delta E$ is a Gaussian distribution for small δE .

- 4.6** Consider N ionized atoms of a monatomic gas with charge q and mass m placed in a cube of side length L , held at temperature T and subject to a uniform electric field $\mathcal{E}$ in the x direction. Ignore interactions between the ions.

- (a) Treat the ions as a classical gas with energies

$$E = \frac{p^2}{2m} - q\mathcal{E}x,$$

and let $x \in [-L/2, L/2]$. Calculate the one-particle partition function Z_1 and hence show that the free energy for such a gas takes the form

$$F = -Nk_B T \ln \left[\frac{n_Q}{n} \frac{L_\phi}{L} \sinh \left(\frac{L}{L_\phi} \right) \right] - Nk_B T,$$

where n is the concentration and

$$n_Q = \left(\frac{mk_B T}{2\pi\hbar^2} \right)^{\frac{3}{2}}$$

is the quantum concentration. Specify L_ϕ in terms of β , q and $\mathcal{E}$.

- (b) Using your answer to part (a) or otherwise, show that the average energy U of the gas is

$$U = Nk_B T \left[\frac{5}{2} - \frac{L}{L_\phi} \coth \left(\frac{L}{L_\phi} \right) \right].$$

- (c) Find the limiting forms of the expression for U in part (b) in the high-temperature ($k_B T \gg q\mathcal{E}L$) and low-temperature ($k_B T \ll q\mathcal{E}L$) limits and comment briefly on your results.

- 4.7** Use the equipartition theorem to estimate the typical angular velocity of a nitrogen molecule at room temperature. (The N–N bond length is 1.0975 Å and the mass of an N atom is 14.0067 amu.)

- 4.8** Consider ultra-cold dilute neutrons, at temperature $T = 1$ mK, which are initially placed at the bottom of a box of height H . How high does the box need to be to ensure that a neutron with average energy remains in the box?

- 4.9** Consider a diatomic ideal gas with N particles in three dimensions at a temperature T , where vibrations and internal electronic transitions are unimportant. Calculate the heat capacity at constant volume, C_V , by first finding the partition function for the gas. You may assume that the single-particle partition function for rotations is

$$Z_{\text{rotations}} \simeq \frac{T}{\theta_r},$$

where θ_r is a characteristic temperature for rotations. How would your result change if you considered a monatomic ideal gas?

- 4.10** In an ideal gas the particles are assumed to be non-interacting. For some real gases, a reasonable approximation to the canonical partition function for a gas with N particles at temperature T in volume V is

$$Z = c \left(\frac{V - Nb}{N} \right)^N \left(\frac{2\pi mk_B T}{\hbar^2} \right)^{\frac{3N}{2}} e^{\frac{N^2 a}{V k_B T}},$$

where a , b and c are positive constants. Show that this partition function implies the van der Waals equation of state

$$(P + n^2 a)(1 - nb) = \frac{Nk_B T}{V},$$

where $n = N/V$.

- 4.11** Suppose we have a lattice of N indistinguishable atoms ($N \gg 1$) in one of three states with energies 0 , ϵ and 4ϵ .
- (a) If the mean energy per atom is ϵ , what is the temperature?
- (b) Find the entropy as a function of temperature at this energy.
- 4.12** Consider dilute He gas in a three-dimensional box with the initial condition that all atoms have a speed of 1000 m s^{-1} but are moving in random directions. After some time the gas will equilibrate. What will be the temperature of the gas at equilibrium (neglect the exchange of heat between the gas and the environment)?
- 4.13** Calculate the entropy S and energy U for a quantum harmonic oscillator and hence determine the heat capacity at constant volume, C_V .
- 4.14** Calculate the entropy S and energy U for a classical harmonic oscillator and hence determine the heat capacity at constant volume, C_V . Comment on whether your answer agrees with what you expect from the equipartition theorem.
- 4.15** Suppose that a system of N atoms of type A is placed in diffusive contact with a system of N atoms of type B at the same temperature and volume. Show that after diffusive equilibrium is reached, the total entropy has increased by $2N \ln 2$ compared to the entropy when the systems were not in contact. This entropy increase is known as the entropy of mixing. If the atoms are identical ($A = B$), show that there is no increase in entropy when diffusive contact is established.
- 4.16** Find the partition function for a symmetric rotor in the classical limit. Hence compare the heat capacity of a gas of molecules that can be modelled as symmetric rotors with a gas of molecules that can be modelled as linear rotors. Discuss your result with reference to the equipartition theorem.
- 4.17** Consider non-interacting spins as we did in [Section 2.4](#). The energy for each spin s is given by

$$\epsilon = -s\mu B,$$

where the spin s can take values ± 1 . For a system of N spins in a solid, find the Helmholtz free energy.

- (a) Using the result that the magnetization is

$$M = -\frac{1}{\mu} \frac{\partial F}{\partial B},$$

find the magnetization per spin, m , and show that it agrees with the result we found in [Eq. \(2.36\)](#).

- (b) Find the heat capacity at constant magnetic field and show that it displays a Schottky anomaly as a function of temperature.

- 4.18** Deuterium hydride (HD) has a boiling point of 22.1 K and characteristic temperatures $\theta_r = 64 \text{ K}$ and $\theta_v = 5518 \text{ K}$ for rotations and vibrations, respectively. At atmospheric pressure and a temperature of 3000 K the gas starts to dissociate.

Treating HD molecules as a linear rotor, make a plot of the heat capacity over the temperature range 22.1 K to 3000 K.

- 4.19** Hydrogen gas (H_2) can exist in two slightly different forms, ortho and para hydrogen, depending on the state of the spins of the hydrogen nuclei. Treating hydrogen as a linear rotor, the rotational energy is

$$\epsilon_J = \frac{\hbar^2 J(J+1)}{2I}.$$

For ortho hydrogen the spins are in a triplet state, for which the degeneracy is 3 and J can only take even integer values 0, 2, 4, ..., while for para hydrogen the spins are in a singlet state, for which the degeneracy is 1 and J can only take odd integer values 1, 3, 5, Noting that the proton–proton bond distance in hydrogen is 0.747 Å, find the ratio of ortho to para hydrogen at (i) a temperature of 25 K and (ii) high temperatures.

You may ignore non-rotational degrees of freedom since these are the same for both species.

- 4.20** Solve [Problem 1.9](#) using the canonical ensemble by recalling that the energy of the polymer chain is given by

$$E = \sum_i \frac{p_i^2}{2m} - \mathbf{X} \cdot \mathbf{R},$$

where $\mathbf{X}$ is the tension applied to the chain and $\mathbf{R} = \sum_i \mathbf{r}_i$ is the extension of the chain. Parametrize the orientation of the i^{th} monomer by a vector written in spherical polar co-ordinates:

$$\mathbf{r}_i = a (\cos \phi_i \sin \theta_i, \sin \phi_i \sin \theta_i, \cos \theta_i).$$

Assuming that $\mathbf{X}$ is oriented along the z direction, show that the z component of the average displacement is

$$\langle R_z \rangle = \frac{Na}{\sinh(\beta X a)} \left[\cosh(\beta X a) - \frac{\sinh(\beta X a)}{\beta X a} \right].$$

Specify under what circumstances we may consider the tension as small, and show that in this limit the polymer may be regarded as a spring with spring constant

$$k = \frac{3k_B T}{Na^2}.$$

- 4.21** Consider an ideal gas confined in a potential

$$V(x, y, z) = \alpha(x^n + y^n + z^n),$$

where n is a positive even integer. Calculate the classical partition function and hence show that the heat capacity is

$$C_V = \left(\frac{3}{2} + \frac{3}{n} \right) Nk_B.$$